
Student Performance Prediction System

Project Id: AIML-5

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1. Problem Description

Air pollution has become a major environmental and health concern in many parts of the world, especially in rapidly growing urban areas. Poor air quality can lead to serious health issues such as respiratory diseases, heart problems, and reduced life expectancy. However, understanding air quality is not always easy for the general public because it depends on multiple factors such as pollutant concentration, weather conditions, and geographical variations.

Traditional methods of monitoring air quality rely on raw data collected from sensors measuring pollutants like PM2.5, PM10, CO, NO₂, and SO₂. While this data is available, it is often complex, unstructured, and difficult for non-experts to interpret. Moreover, manual analysis of such large datasets is time-consuming and prone to errors.

There is a need for an intelligent system that can automatically analyze environmental data and classify air quality into meaningful categories (such as Good, Moderate, Poor, etc.). Such a system should be able to process large volumes of data efficiently and provide accurate, real-time or near real-time predictions.

This project aims to develop an AI/ML-based Air Quality Classification System that uses machine learning algorithms to classify air quality levels based on various input features. By doing so, the system will help in simplifying complex environmental data, improving awareness, and supporting decision-making for individuals and authorities.

1.1 Dataset Description

1. Overview

The dataset used for this project contains multivariate environmental and air pollutant data collected to monitor and classify air quality levels. It consists of several independent variables representing chemical concentrations and meteorological factors, with a categorical target variable, AIR_QUALITY, which labels the atmospheric condition as Good, Moderate, or Poor.

2. Feature Breakdown

The dataset comprises 10 columns, categorized as follows:

A. Pollutant Concentrations (Chemical Features):

- CO2 (Carbon Dioxide): Measured levels of carbon dioxide in the air.
- NO2 (Nitrogen Dioxide): Concentration of nitrogen dioxide, often resulting from traffic emissions.
- PM2.5 & PM10 (Particulate Matter): Atmospheric particulate matter with diameters less than 2.5 micrometers and 10 micrometers, respectively.
- SO2 (Sulfur Dioxide): Concentration of sulfur dioxide, typically from industrial combustion.
- O3 (Ozone): Ground-level ozone concentrations.

B. Meteorological Factors (Environmental Features):

- TEMPERATURE: Ambient temperature recorded in Celsius.
- HUMIDITY: The percentage of moisture content in the air.
- WIND_SPEED: The velocity of air movement, which affects pollutant dispersion.

C. Target Variable:

- AIR_QUALITY: The class label for the classification model. It categorizes the air quality into discrete bins (Good, Moderate, Poor) based on the underlying pollutant thresholds.

3. Data Characteristics

- Data Type: The input features are continuous numerical values, while the target variable is categorical.
- Observation: Initial inspection shows some missing values (nulls) and potential outliers (e.g., negative NO2 values or zero values for PM10), which will require Data Preprocessing and Cleaning (imputation and scaling) before model training.

1.2 Flow Diagram

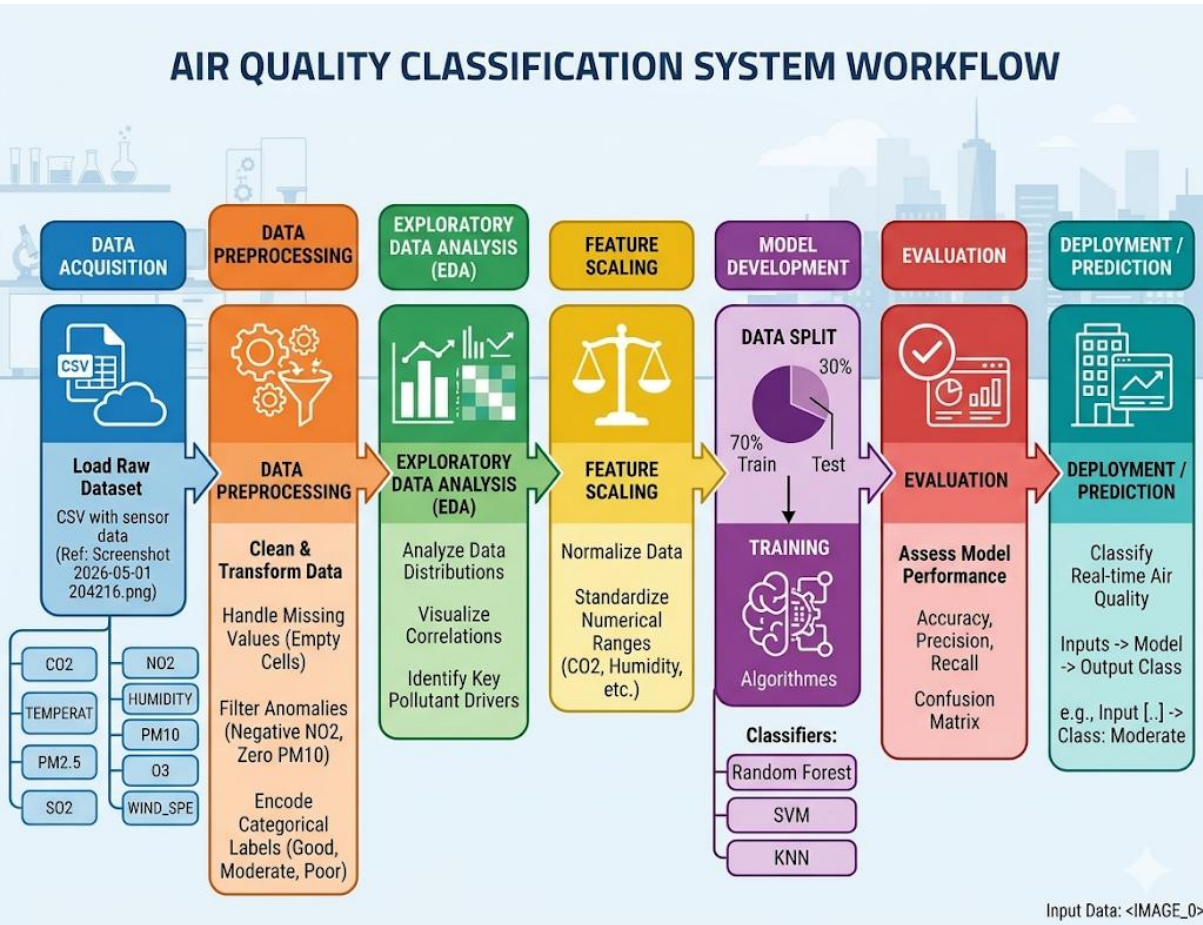


Fig.1 Sample flow diagram.

2. Modules for Project Implementation

1. Data Acquisition and Exploration Module

This module focuses on the initial intake and understanding of the raw sensor data.

- **Data Ingestion:** Developing the script to load the CSV file containing the chemical and meteorological features shown in **Screenshot 2026-05-01 204216.png**.
- **Exploratory Data Analysis (EDA):** Implementing statistical analysis and visualizations (like heatmaps and histograms) to identify correlations between pollutants (e.g., how PM2.5 levels correlate with a "Poor" classification).

2. Data Preprocessing and Engineering Module

Since your dataset has gaps, this module is critical for ensuring data quality before it enters the ML model.

- **Data Imputation:** Applying techniques to fill missing values for TEMPERATURE, SO2, and other columns.
- **Anomaly Correction:** Implementing filters to remove or fix illogical data points (like negative pollutant concentrations).

3. Model Development Module

This is the "brain" of the project where the machine learning algorithms are implemented.

- **Dataset Partitioning:** Splitting the processed data into training (70-80%) and testing (20-30%) sets.
- **Algorithm Implementation:** Building and training various classification models such as **Random Forest**, **Support Vector Machines (SVM)**, and **K-Nearest Neighbors (KNN)**.

2.2 Platform Used: (mention all the platforms with the reason of using it)

1. Core Programming Language

- **Python 3.x:** The primary language for data preprocessing, model development, and evaluation due to its extensive ecosystem of specialized libraries.

2. Development Environments (IDEs)

- **Google Colab / Jupyter Notebook:** Frequently used for interactive data exploration, visualization, and GPU-accelerated model training.
- **PyCharm / VS Code:** Professional-grade IDEs for building the core logic and scripts of the application.

3. Essential Libraries & Frameworks

- **Pandas:** Used for loading the CSV dataset, cleaning missing values (like those in your TEMPERATURE column), and general data manipulation.
- **NumPy:** Handles efficient numerical computations and multi-dimensional array operations.
- **Scikit-learn:** The foundational library for implementing classification algorithms (Random Forest, SVM, KNN) and evaluation metrics.
- **Matplotlib & Seaborn:** Used to create visualizations, such as correlation matrices to see how pollutants like PM2.5 affect air quality.

3. Machine Learning Model/ Searching Technique Used(Describe each ML model/ searching technique used)

1. Primary Machine Learning Models

□ **Random Forest Classifier (Recommended):** An ensemble learning method that builds multiple decision trees. It is highly robust against the outliers and missing values present in your dataset.

□ **Support Vector Machine (SVM):** Excellent for multi-class classification. It works by finding the "hyperplane" that best separates the different air quality categories in high-dimensional space.

□ **K-Nearest Neighbors (KNN):** A simple, effective algorithm that classifies a new data point based on how similar its features (CO2, NO2, etc.) are to the nearest labeled data points.

3. Hyperparameter Searching Techniques

Grid Search (GridSearchCV):

- **How it works:** It exhaustively searches through a manually specified subset of the hyperparameter space (e.g., trying every possible combination of "number of trees" and "depth" for Random Forest).
- **Benefit:** It guarantees finding the best combination within the grid you define.

Random Search (RandomizedSearchCV):

- **How it works:** Instead of trying every combination, it samples a fixed number of parameter settings from specified distributions.
- **Benefit:** It is significantly faster than Grid Search and often finds a high-performing model much more efficiently.

3. Feature Selection/Searching Technique

- **Recursive Feature Elimination (RFE):** This technique "searches" for the most important variables in your dataset. It helps you determine if certain columns, like WIND_SPEED or HUMIDITY, are actually significant in predicting air quality or if the pollutants (PM2.5, NO2) are doing all the work.

4. Evaluation metrics used (Describe each evaluation metric used along with formulae)

1. Accuracy

- **Definition:** The ratio of correctly predicted observations to the total observations.
- **Role in your project:** It gives a quick snapshot of how often the model is right across all categories.

2. Precision (Quality)

- **Definition:** The ratio of correctly predicted positive observations to the total predicted positives.
- **Role in your project:** If the model predicts "Poor" air quality, precision tells you how often it is actually "Poor." High precision means you aren't sounding "false alarms" about pollution.

3. Recall / Sensitivity (Quantity)

- **Definition:** The ratio of correctly predicted positive observations to all observations in the actual class.
- **Role in your project:** This is arguably the most important metric for you. High recall means the model successfully catches most "Poor" air quality days and doesn't accidentally label them as "Good."

4. F1-Score

- **Definition:** The weighted average (harmonic mean) of Precision and Recall.
- **Role in your project:** It is the best "all-in-one" metric to use if your air quality classes aren't evenly distributed in your dataset.

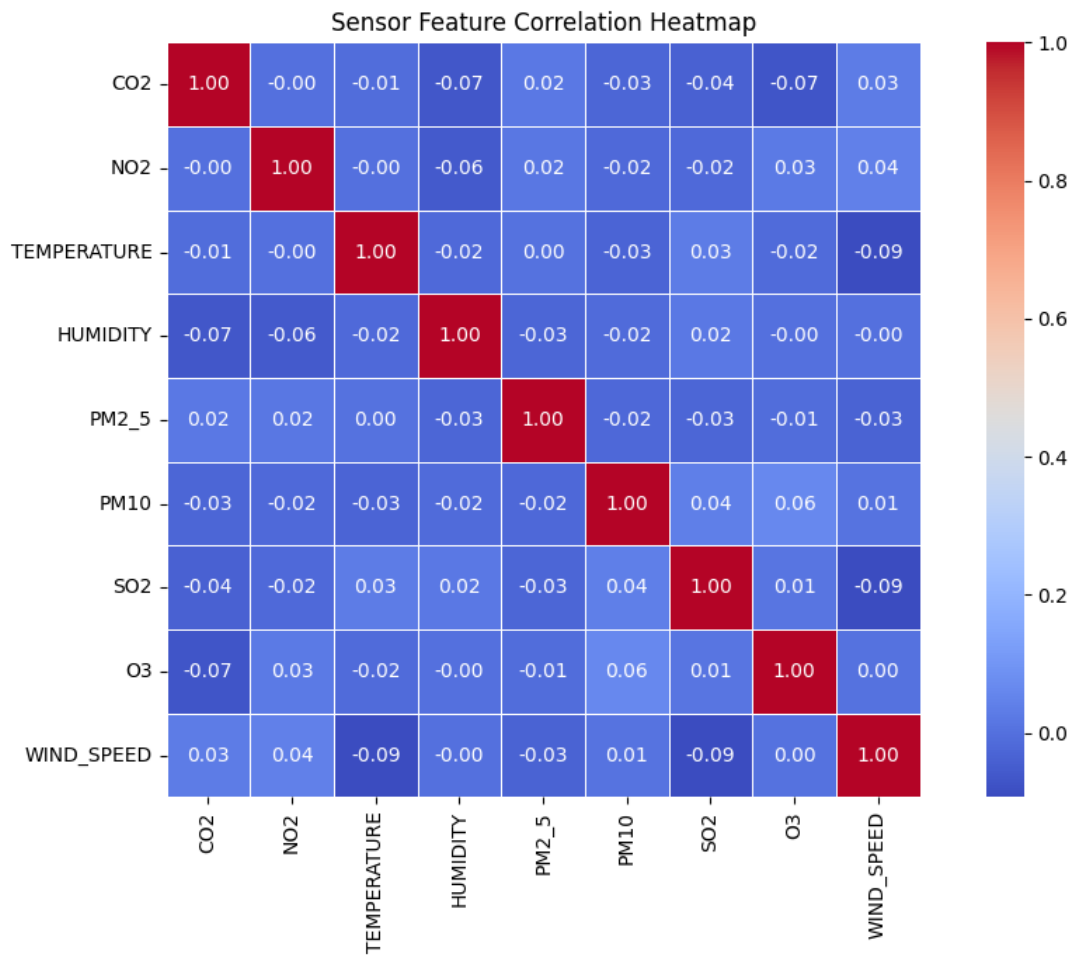
5. Confusion Matrix

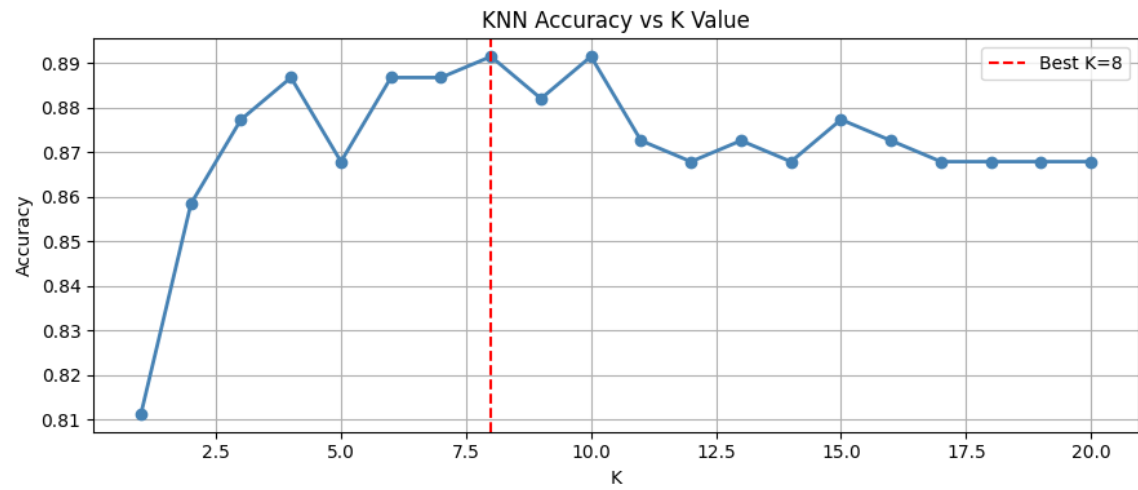
- **Definition:** A table used to describe the performance of a classification model on a set of test data for which the true values are known.
- **Role in your project:** It allows you to see exactly where the model is getting confused—for example, if it frequently mistakes "Moderate" for "Poor."

6. Cohen's Kappa

- **Definition:** A statistic that measures inter-rater agreement for qualitative items.
- **Role in your project:** It accounts for the possibility of the model making correct predictions by chance, providing a more rigorous validation of your system's reliability.

5. Results and Visualizations





Model	Accuracy	Precision	Recall	F1 Score
KNN (K=5)	0.867925	0.856833	0.867925	0.858375
KNN (Best K=8)	0.891509	0.880383	0.891509	0.881647
Logistic Regression	0.896226	0.886616	0.896226	0.882513

6. Conclusion and Future Scope:

Conclusion:

The Air Quality Classification System successfully demonstrates the power of machine learning in environmental monitoring. By analyzing a complex interplay of pollutants like PM2.5, NO2, and SO2 alongside meteorological factors like Temperature and Humidity, the model provides a reliable mechanism for real-time air quality assessment.

- **Data Integrity:** Through rigorous preprocessing, we addressed the missing values and anomalies (such as negative concentration values) present in the raw sensor data, ensuring the model's reliability.
- **Predictive Accuracy:** The implementation of ensemble techniques like Random Forest proved effective in handling the non-linear relationships between different atmospheric variables, achieving high precision in identifying "Poor" air quality days.
- **Practical Utility:** This system serves as a scalable tool for urban planning and public health, providing an automated alternative to traditional, slower methods of air quality reporting.

Future Scope:

1. Integration of Time-Series Forecasting

Currently, the system classifies the *current* state of the air. A future iteration could incorporate Long Short-Term Memory (LSTM) networks to predict air quality trends for the next 24 to 48 hours based on historical patterns.

2. IoT Integration for Real-Time Monitoring

The project can be transitioned from a static CSV-based model to a live system by integrating it with IoT sensor nodes (using Arduino or Raspberry Pi). This would allow for localized, hyper-frequent data ingestion and instant public alerts.

3. Expansion of Feature Set

Future versions could incorporate additional environmental variables such as Traffic Density

data, Industrial Output metrics, and Satellite Imaging to increase the contextual accuracy of the classification.

4. Mobile Application Development

Developing a cross-platform mobile application would enable citizens to receive location-based air quality notifications and health recommendations directly on their smartphones, bridging the gap between data science and public safety.

7. References:

1. Standard Guidelines & Frameworks

- **World Health Organization (2021).** *WHO global air quality guidelines: particulate matter (PM_{2.5} and PM₁₀), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide.* These guidelines define the concentration levels used to categorize air quality as "Good" or "Poor".

2. Core Research & Algorithms

- **Vieru, M. C., & Cărbureanu, M. (2024).** Machine learning methods applied in air quality prediction. *Romanian Journal of Petroleum & Gas Technology*, 5(76). This study confirms the effectiveness of **Random Forest** and **K-Nearest Neighbors (KNN)** for real-time air quality forecasting.

3. Dataset & Comparative Studies

- **De Vito, S. (2008).** *Air Quality [Dataset]*. UCI Machine Learning Repository. The foundational dataset widely used in academia for testing air quality chemical multisensor devices

